

①

```
class person
```

```
    // properties
```

```
    character name
```

```
    integer Age
```

```
    constructor (name, age)
```

```
        this.name = name
```

```
        this.age = age
```

```
    End constructor
```

```
End class
```

```
class students inherits person
```

```
    int registernumber
```

```
    Array of integer marks[5]
```

```
    constructor (name, age, regnum, Array marks)
```

```
        super(name, age)
```

```
        This.reg = reg;
```

```
        This.marks = marksArray
```

```
    End constructor
```

```
    method calculateAverage()
```

```
        sum = sum + mark
```

```
    End For
```

```
        Return sum/5
```

```
    end method
```

```
class ResearchStudent inherits student
```

```
    character Researchareq
```

```
    constructor (name, age, reg, marksArray)
```

```
        super (name, age, regnum, marksArray)
```

End constructor

method displayCompleteDetails()

End method

End class

//START main execution

Array studentMarks = [85, 90, 78, 92, 88]

ResearchStudent obj = new ResearchStudent("

John Doe", 24, "Reg123", marks, "A1")

obj.displayCompleteDetails()

End

②

BEGIN

READ N

READ ARRAY

FOR i ← 0 TO N-1

FOR j ← i+1 TO N-1

IF A[i] == A[j]

FOR k FROM j TO N-2

A[k] = A[k+1]

END FOR

N = N-1

j = j-1

end if

end FOR

end FOR

```

③ class Area
    method area(int side)
        Return side * side;
    End method
method area (int length, int breadth)
    Return length * breadth;
End method
end class
Task

```

④ output:

```

sum = 0
iteration 1 (i=1)
    1 mod 2 == 0 false.

    sum = sum - i
        = 0 - 1
    current sum = -1
iteration 2 (i=2)
    2 mod 2 == 0
    sum = sum + i
    current sum = -1 + 2 = 1
iteration 3 (i=3)
    3 mod 2 == 0

```

sum = sum - i

1 - 3

sum = -2

iteration 4 (i=4):

4 mod 2 == 0

sum = sum + i

sum = 2

iteration 5 (i=5)

sum = sum - i

= 2 - 5

= -3

Final result: -3

5

START

largest = arr[0]

second-largest = arr[1]

if n < 2 then

System.out.println("invalid input")

for i from 2 to n-1

if arr[i] > largest

second-largest = largest

largest = num

else if arr[i] > second-largest and

arr[i] != largest.

second-largest = num

end if

end for

```
End if  
end for
```

⑥

```
interface paymentmethod
```

```
    method pay(amount: Decimal)
```

```
End interface
```

```
class creditcard implements paymentmethod
```

```
    method pay (amount: decimal)
```

```
        print "processing credit card"
```

```
    End method
```

```
end class
```

```
class upi implements paymentmethod
```

```
    method pay (amount: decimal)
```

```
        print "processing upi payment"
```

```
    end method
```

```
end class
```

```
class netBanking implements paymentmethod
```

```
    method pay (amount: Decimal)
```

```
        print "processing net Banking"
```

```
    end method
```

```
end class
```

```
declare currentPayment: payment method
```

```
declare userchoice : integer
```

```
declare paymentAmount: decimal
```

```
Enter amount to pay
```

```
select payment method
```

Switch user case.

case 1:

current payment = new creditCard()

case 2:

current payment = new upi()

case 3:

current payment = new netBanking()

default

print "invalid"

exit

current payment . pay (payment amount)

END

①

class Animal

method speak()

print Animal

End method

End class

class Dog extends Animal

method speak()

print Barks

End method

End class

Animal obj = new Dog

obj . speak()

②

```
class employee
```

```
  DATA
```

```
    id
```

```
    name
```

```
  constructor (id, name)
```

```
    this.id = id
```

```
    this.name = name
```

```
  End constructor
```

```
End class
```

③

⑨

START

class vehicle

DATA

vehicle NO

owner name

method readDetails()

input vehicleNO

input owner name

End method

method display()

display vehicleNO, owner name

End method

End class

class car inherits vehicle

DATA

fuel type

method readDetails()

CALL

vehicle.readDetails()

input fueltype

End method

method display()

display car

display vehicleNO, owner name, fuel  
type

End method

End class

class truck inherits vehicle

DATA

load capacity

method readDetails()

call

Vehicle.readDetails()

input load capacity

end method

method display()

display truck

display vehicle no, owner, load capacity

end method

end class

main class

for i = 1 to N

if vehicletype = "car" then

vehicle(i) ← new car

else

new truck

end if

end for

for i = 1 to N

call

vehicles(i).display()

end for

end main

stop